

No.	Solution	Remarks
5(a)(i)	$R_{\text{external}} = \frac{R_x R_y}{R_x + R_y} + R_z$ $= \frac{(4)(2)}{4 + 2} + 2$ $= 3.33 \, \Omega$ Terminal p.d. $= \frac{R_{\text{external}}}{R_{\text{external}} + 0.5} (\text{e.m.f.})$ $= \frac{3.33}{3.33 + 0.5} (3.0)$ $= 2.608$ $\approx 2.6 \, \text{V}$	[1] [1]
5(a)(ii)	X, Z, Y	[1]
5(b)(i)	By potential divider principle, $V_{\text{AB}} = \frac{R_{\text{AB}}}{R_{\text{AB}} + 1} (6.0)$ $= \frac{2.0}{2.0 + 1.0} (6.0)$ $= 4.0 \, \text{V}$	[1]
5(b)(ii)	$\frac{L_{\text{AC}}}{L_{\text{AB}}} = \frac{V_{\text{LP}}}{V_{\text{AB}}}$ $L_{\text{AC}} = \frac{V_{\text{LP}}}{V_{\text{AB}}} \times L_{\text{AB}}$ $= \frac{2.61}{4.0} \times 100$ $= 65 \, \text{cm}$	[1] correct working [1] correct answer
5(b)(iii)	Effective resistance across MN decreases causing current in lower circuit to increase. Terminal p.d. across cell decreases. Shorter balance length.	[1] explain [1] correct conclusion

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